DECENTRALIZED CLEAN ENERGY INFRASTRUCTURE

EcoBin

"Throw Waste. Create Energy."

AI waste sorting + on-site biogas for campuses & societies

25-30	>98%	Rs.1L+
LPG cylinders saved/month	AI sorting accuracy	saved monthly per campus

echobin.vercel.app | contact@ecobin.tech | Autonomous Bio-Utility



EcoBin Autonomous Segregation Node

THE PROBLEM

Institutions Pay Lakhs Monthly for a Broken Waste System

Mixed waste. Failed sorting. Diesel trucks. Landfill fines.

1.5 Lakh MT

municipal solid waste
generated daily in India

>40%

human sorting
error rate at source

Rs. 1.5L/mo

average haulage bill
paid by campuses

HOW THE BROKEN SYSTEM WORKS TODAY

Mixed Disposal

Zero sorting
at source

->

Manual Bins Fail

>40% error,
dirty loads

->

Diesel Haulage

Rs. 80k/mo to haul
70% water

->

Landfill + Fines

Rs. 50k NGT
fines





MARKET TIMING

Why Now? Four Tailwinds Make This the Right Moment

Regulations. Cost spikes. Cheap AI sensors. ESG pressure — all converging.

Regulation Mandates Action

SWM 2016 legally requires on-site treatment for all bulk generators >100 kg/day.
Rs. 50,000 NGT fines now actively enforced.

Diesel & LPG Costs Surged 40%

Haulage costs up 40% in 3 years.
LPG cylinder prices at all-time high.
Institutions urgently seeking alternatives.

AI Sensors Now Cost <Rs. 800

ESP32-S3 AI camera modules: <Rs. 800.
Edge inference viable in Rs. 3L hardware.
Moore's Law finally reached waste-tech.

ESG & NAAC Now Mandatory

Universities need NAAC Green credentials.
Corporates face SEBI BRSR net-zero mandates.
Green infra is a procurement priority.

THE SOLUTION

Drop Any Waste. Get Free Cooking Gas.

Fully autonomous 4-step closed loop. Zero manual sorting. Zero diesel trucks.

01

Drop Waste

User drops anything.
No sorting needed.

->

02

AI Vision Sort

<150ms edge AI
sorts dry vs wet

->

03

Bio-Digestion

Organic food waste
becomes clean biogas

->

04

Free Cooking Gas

Piped directly to
kitchen stoves

BEFORE vs. AFTER

BEFORE

Mixed Waste -> 40% Error -> Rs.80k Hauling -> Landfill + Rs.50k Fine

AFTER (EcoBin)

Single Drop -> AI Sorts -> Biogas to Kitchen -> Rs.1L+ Saved Monthly

THE PRODUCT

EcoBin Hardware: Plug-and-Play Smart Waste Node

Installs in 3-5 days. No civil work. Modular for 100-500 kg/day.



EcoBin Autonomous Hardware Node | 100–500 kg/day Modular Capacity

AI Vision Sort

ESP32-S3 + YOLOv8
<150ms | >98% accuracy

Bio-Digestion

Mesophilic 37C dome
Auto pH anti-souring

Free Cooking Gas

65% CH4 purity
Piped to kitchen stoves

Clean Recyclables

Dry bales: PET, paper
Sold to industrial mills

MARKET OPPORTUNITY

Rs. 14,200 Cr Market — 1.2 Lakh Generators Legally Mandated to Act

Bottom-up sizing. Every bulk waste generator >100 kg/day must comply with SWM 2016.

TAM **Rs. 14,200 Cr**

1,20,000 bulk waste generators
across urban India

120k x Rs.1.18L avg

SAM **Rs. 3,200 Cr**

25,000 universities & gated societies
in Tier-1 cities

25k x Rs.1.28L avg

SOM **Rs. 95 Cr**

800 beachhead deployments
Delhi-NCR & Bengaluru (3-yr)

800 hubs x Rs.12L ARR

GO-TO-MARKET EXECUTION ROADMAP

Stage 1
Month 0-6

Prototype Build
& Pilot Deployment

->

Stage 2
Month 6-12

20 university
campuses (NCR)

->

Stage 3
Month 12-24

150 gated societies
JLL/CBRE channel

->

Stage 4
Year 2+

800 nodes
Series A ready

ARCHITECTURE & PILOT TARGETS

Engineered Unit Capacity & Projected Savings

Target operational specifications designed for standard higher-ed campuses:

250 kg

food waste
daily capacity target

28 LPG

cylinders replaced
projected per month

>95%

machine vision
sorting precision target

Rs. 1L+

net savings per
campus per month (est.)

KEY ENGINEERING & PILOT TARGETS

Dual-stream automated diverter: dry recyclables to baler, food scraps to macerator

Mesophilic 37°C anaerobic reactor with automated pH buffering against acid souring

Direct methane piping (65% CH4) designed to replace commercial 19kg mess LPG cylinders

Sub-150ms sensor fusion (Sony IMX vision, dielectric moisture, inductive metal)

100% compliant with Ministry of Environment SWM 2016 bulk generator mandate

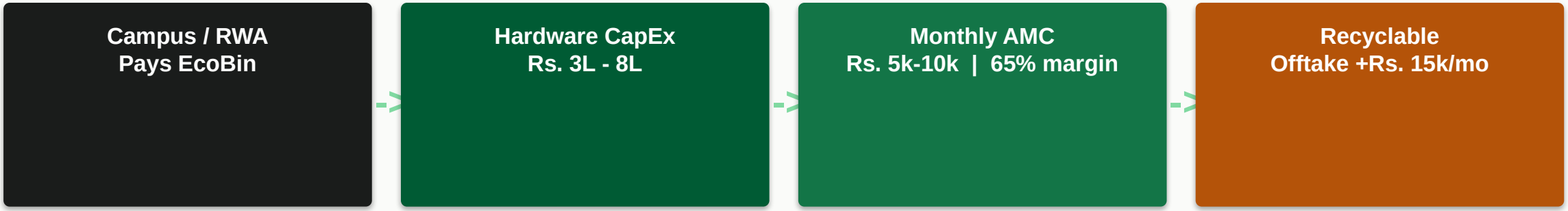
Source: EcoBin Systems Architecture & Pilot Engineering Benchmarks

BUSINESS MODEL & UNIT ECONOMICS

Hardware Sale + Sticky Monthly Retainer = 16-20 Month Payback

Customer recovers full capital cost in <20 months purely from operational savings.

HOW ECOBIN MAKES MONEY



CAMPUS UNIT ECONOMICS (250 kg/day)

Hauling Contracts Eliminated	Rs. 45,000 / mo
LPG Cylinders Replaced (28 nos)	Rs. 56,000 / mo
Recyclable Sales	+ Rs. 6,500 / mo
EcoBin AMC Fee	- Rs. 7,500 / mo
NET MONTHLY SAVING	**Rs. 1,00,000 / mo**
Capital Payback Period	**16-20 months**

38-42%
Hardware Margin

65-70%
AMC Margin

>6.8x
LTV / CAC

COMPETITION & MOAT

No One Combines AI Sorting + On-Site Biogas. EcoBin Owns This.

Competitors solve one piece. EcoBin solves the complete system.

Feature	EcoBin	Manual 2-Bin	Compost Pit	Old Biogas
AI Auto-Sorting	YES <150ms	NO Manual	NO None	NO None
On-Site Biogas	YES	NO	NO	RISK Unstable
Zero Human Sorting	YES	NO 40% fail	NO Manual	NO Manual
SWM 2016 Compliant	YES Full	PART	PART	PART
Rs.1L+/mo Saving	YES	NO	NO	NO
Anti-Souring Control	YES Auto	NO	NO	NO Manual

OUR 4 COMPETITIVE MOATS

Proprietary AI Dataset

20k+ real waste images, growing with every node

Anti-Souring Bioprocess IP

Auto pH dosing loop — no competitor has this

Utility Lock-In

Gas pipes = sticky. <1.5% annual churn

Regulatory Alignment

100% SWM 2016 compliant on-site bio-utility design

THE TEAM & INVESTMENT ASK

Raising Rs. 1.5 Cr Seed to Deploy Across 20 Target Campuses

Domain-first team. Clear milestones. Every rupee de-risks a specific technical risk.

TEAM CAPABILITIES

Embedded Systems & Firmware

ESP32-S3 | FreeRTOS | OTA | Edge AI

Computer Vision & ML

YOLOv8 | TFLite | Custom 20k Dataset

Anaerobic Bio-Process Eng.

Biogas | VFA Monitoring | pH Control

Institutional B2B Sales

Campus Admin | SWM Compliance | GTM

Hardware & Manufacturing

Fabrication | Modular Design | BoM

MILESTONE LADDER — USE OF FUNDS

RAISING: Rs. 1.5 Cr SEED ROUND

12-month runway | 20 campuses | Rs. 1.2 Cr ARR

M1 — **Rs. 35L** — Month 0-3

Gen-2 hardware finalized
BIS certifications done
Risk: scaling proven

M2 — **Rs. 70L** — Month 3-9

20 campus hubs deployed
3-person sales team hired
Target: Rs. 12L ARR

M3 — **Rs. 45L** — Month 9-18

NCR assembly hub opened
JLL/CBRE channel signed
Target: 100 nodes, Series A

DECENTRALIZED URBAN BIO-UTILITY | SEED STAGE

Thank You

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Open for institutional pilots, municipal partnerships & strategic collaborations